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Dmitry Pashchenko ; Maxim Nikitin

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Dmitry Pashchenko[1,2,a] and Maxim Nikitin[2]

AFFILIATIONS

[1]Faculty of Mechanical Engineering, Technion—Israel Institute of Technology, Technion, Haifa 3200003, Israel
[2]Department of Mechanical Engineering (Robotics) and MATEC, Guangdong Technion Israel Institute of Technology, 241 Daxue Rd., Guangdong, Shantou 515063, China

[a]Author to whom correspondence should be addressed: dmitry.pa@technion.ac.il and dmitry.pashchenko@gtiit.edu.cn

ABSTRACT

This study investigates the influence of surface dimpling on pressure drop in a confined monolayer of spheres under low diameter-ratio conditions ($D_{\text{tube}}/d_{\text{sphere}} \approx 1.17$). An integrated experimental and computational fluid dynamics approach is employed to analyze the hydraulic performance of smooth and dimpled spheres arranged in a single row within a horizontal pipe, across a range of inter-particle spacing ratios ($\delta/D_p = 1.0$–2.0) and superficial velocities ($U_s = 2$–16 m/s). Experimental results reveal that dimpled spheres consistently exhibit higher pressure drops than smooth spheres, with the relative increase quantified by the ratio $\Psi = (\Delta P/L)_{\text{dimpled}}/(\Delta P/L)_{\text{smooth}}$. This ratio shows a non-monotonic dependence on U_s, reaching a minimum of approximately 1.02 near $U_s = 9$–10 m/s, and is systematically reduced by increasing δ/D_p. Normalization of pressure drop per particle ($\Delta P/N_p$) collapses data onto single power-law trends for each surface type, indicating that individual particle contributions are largely independent of packing density. The increased resistance for dimpled spheres is attributed to enhanced surface area, promoted flow separation, and earlier boundary layer transition. These findings demonstrate that surface texture significantly alters flow resistance in confined arrays, where wall effects and particle proximity dominate, offering insights for applications in filtration, catalytic beds, and narrow-channel flows.

I. INTRODUCTION

The aerodynamic performance of spherical bodies in confined flows is of significant interest across a range of engineering and scientific disciplines, from granular transport and packed bed reactors to sports aerodynamics. In particular, the role of surface texture—such as the dimples found on golf balls—in altering flow behavior and drag has been well-documented in external, high-Reynolds-number flows.[1,2] Dimples are known to promote boundary layer transition and delay flow separation, thereby reducing aerodynamic drag in free-flight conditions.[1,3] However, the influence of such surface modifications on flow resistance in internally confined and ordered configurations remains a subject of ongoing research.

In industrial applications, such as filtration, catalytic beds, and pneumatic conveying, spherical particles, are often arranged in ordered or random packings within ducts or pipes.[4,5] In these systems, the ratio of pipe diameter to particle diameter $D_{\text{tube}}/d_{\text{sphere}}$ is a critical parameter governing flow behavior and transport phenomena.[6,7] When this ratio approaches unity, wall effects dominate, interstitial flow diminishes, and conventional packed-bed correlations—which assume $D/d_p \gg 1$—become invalid.[8,9] Under such conditions, the flow resembles that around isolated bodies in close proximity to walls and neighboring particles, leading to complex interactions between wakes, boundary layers, and pressure fields.[9,10]

A considerable body of literature exists on flow through packed beds, with classical studies such as those by Ergun[11] providing widely used semi-empirical correlations for pressure drop in randomly packed spheres.[5,12–14] These models, however, are predicated on the assumption of a bed diameter significantly larger than the particle diameter, and they do not account for ordered arrangements or surface texturing. Recent investigations have explored the effects of confinement and particle arrangement. For example, studies by Bale *et al.*[15] used direct numerical simulation (DNS) to examine mass transfer in packed beds with varying D/d and reported that wall effects diminish considerably only for $D/d \geq 11$. Similarly, Guo *et al.*[13] experimentally characterized pressure drop in slender packed beds ($1 < D/d < 3$) and found that the Ergun equation fails to predict pressure drop accurately in this regime due to significant wall and channeling effects. Their work highlighted that even minor changes in local packing structure can lead to notable differences in pressure drop, underscoring the sensitivity of flow resistance to geometric details at low D/d ratios.

In parallel, the aerodynamics of dimpled spheres in unconfined flow has been extensively studied, largely motivated by sports engineering. Bearman and Harvey[1] first systematically demonstrated the drag-reducing effect of dimples on golf balls, attributing it to induced turbulence and delayed separation. Subsequent research, including simulations by Choi *et al.*,[3] has detailed the mechanisms of vortex generation and skin-friction modification due to surface dimpling. However, these studies almost exclusively consider isolated spheres in free stream or uniform flow, with minimal attention to the effects of nearby boundaries or adjacent bodies. A notable exception is the work of Alam *et al.*,[16] who examined dimpled spheres in tandem arrangements under external flow, finding that drag benefits can diminish or even reverse depending on spacing due to wake interference. To the authors' knowledge, no prior study has systematically investigated the effect of surface dimpling on flow resistance for spheres arranged in a single row inside a pipe, where both wall confinement and periodic particle spacing govern the flow physics.

The structure and mean porosity of packed beds at low D/d ratios are also known to exhibit complex behavior. Guo *et al.*[6] demonstrated that the mean porosity in densely packed beds with small D/d ratios shows rapid oscillations and is highly sensitive to the exact D/d value and packing method. This structural sensitivity directly impacts hydraulic performance, as confirmed by pressure drop experiments.[13] Furthermore, recent numerical studies on heat and mass transfer in packed beds with small D/d ratios[17–19] reveal that pore-scale structure and temperature significantly influence flow distribution and transfer rates. These findings collectively highlight that, in confined systems with low D/d, the global pressure drop and transfer phenomena cannot be predicted by porosity alone; the specific arrangement and surface characteristics of particles play a decisive role.

This study, therefore, addresses a clear gap in the literature by investigating the effect of surface dimpling on flow resistance in a simplified yet representative geometry: a densely packed monolayer of spheres arranged in a single row inside a horizontal pipe with a low diameter ratio ($D_{\mathrm{tube}}/d_{\mathrm{sphere}} \approx 1.17$). The primary objective is to quantify how dimpled surfaces alter the pressure drop compared to smooth surfaces across a range of inter-particle separation distances and superficial velocities. A secondary aim is to develop normalized correlations for pressure drop that account for both wall effects and particle arrangement, extending the understanding of confined multiphase flows beyond traditional packed-bed models.

The work employs an integrated experimental and computational fluid dynamics (CFD) approach. Experiments provide reliable pressure-drop measurements for smooth and dimpled spheres under controlled conditions, while CFD simulations extend the parameter space and offer detailed flow visualization. The results are analyzed in terms of length-based and particle-based normalizations, and a modified friction factor is proposed to characterize flow resistance in this configuration.

Ultimately, this research seeks to bridge the gap between classical aerodynamics of isolated dimpled spheres and practical engineering flows involving confined particle arrays. The findings may inform the design of surfaces for reduced flow resistance in narrow channels, packed columns, and other applications where wall and proximity effects cannot be neglected.

II. METHODOLOGY

The methodology of this study is primarily experimental, aimed at investigating pressure drop across a monolayer of spheres within a confined pipe under varying design and operating conditions. A complementary computational fluid dynamics (CFD) study was conducted to extend the experimental dataset beyond the tested parameter range and to provide detailed flow field insights. The numerical results were verified through mesh dependency analysis, residual convergence checks, and monitoring of key solution variables, and were validated against the experimental data.

A. Experiments

1. Experimental setup

The experimental setup consisted of a horizontal polycarbonate pipe with an inner diameter of 50 mm and a length of 1000 mm (Fig. 1). A 750 W centrifugal blower (CZ-TD750W) forced air through the pipe, capable of a volumetric flow rate of up to 1060 m^3/h at a pressure head of up to 2.3 kPa. The inlet section featured a 100 mm long hub terminating in three rigid stoppers. Solid spherical particles with a diameter D_p were mounted inside the pipe in a single row with a uniform center-to-center spacing δ, which was varied between experimental series. The number of particles n was selected to utilize the entire available test section length. The spheres were placed in contact with the lower pipe wall to ensure mechanical stability and prevent flow-induced vibrations that occurred in preliminary tests with centered suspensions.

The blower was connected via an 80 mm flexible spiral duct. Flow was controlled by a single-phase inverter (AS2-107), adjusting the motor frequency from 10 to 50 Hz in 5 Hz steps. The velocity was measured by an anemometer (DLY-1603A, ±1% FS) with its rotary vane probe installed directly in the blower's outlet hub. The superficial velocity U_s was derived from the measured values, accounting for the effective cross-sectional areas of the vane and the pipe. The pressure drop across the packed bed was measured by a differential pressure sensor (CCY-30, ±0.5% FS) with a full-scale range of 1 kPa.

Two particle types were investigated: smooth, nickel-coated plastic spheres and dimpled golf balls. Both had a diameter of 42.67 mm. The center-to-center distance between adjacent particles was maintained by an ABS separator (a solid rod of diameter 4 mm) fixed to the particles with glued silicone suction cups. A set of separators with lengths of 10, 15, 20, 25, 30, and 35 mm was used. Due to surface specifics, each suction cap added 1.4 mm to a separator for smooth particles and 2.1 mm for dimpled ones. Thus, the actual inter-particle spacing ratios were

$$\delta/D_p = \{1.0, 1.3, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0\}.$$

Both ranges corresponded approximately to $\delta = D_p$ to $2D_p$. The example of monolayers of dimpled and smooth spheres with $\delta = 2D_p$ is presented in Fig. 2.

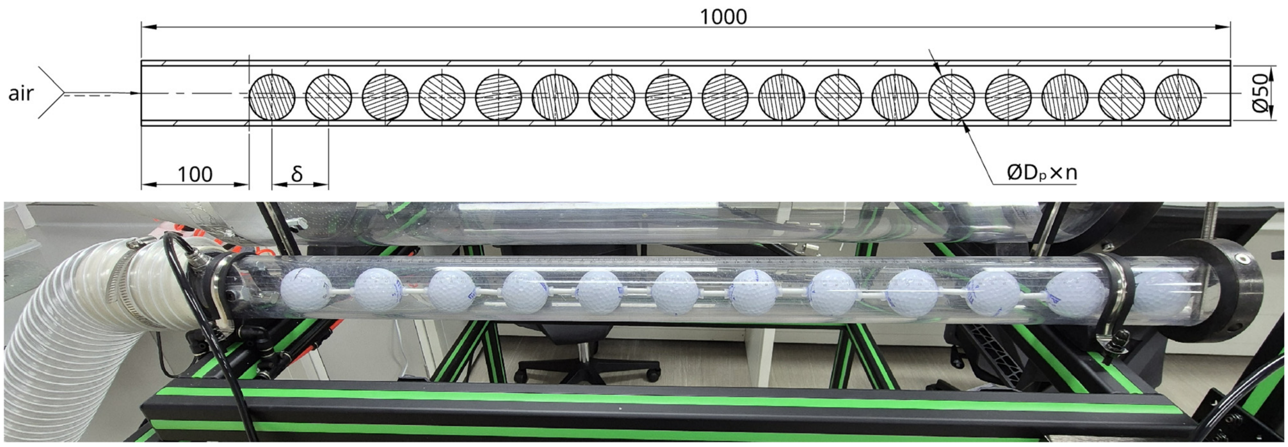

FIG. 1. Schematic of the experimental setup: horizontal polycarbonate pipe (ID 50 mm, length 1000 mm) with a single row of spheres.

2. Experimental procedure and data acquisition

Experimental data were obtained in 14 series with fixed design parameters (particle type, particle count n, and center-to-center distance δ). For each series, the superficial velocity U_s was varied by adjusting the inverter output frequency. Pressure drop ΔP was recorded for each velocity step after flow stabilization.

3. Experimental data validation

Each experimental series was compared with the following analytical correlation to identify outliers:

$$\Delta P/L = A \cdot U_s^2 + B \cdot U_s, \tag{1}$$

where A and B are fitting coefficients. Data points lying outside two standard deviations ($2\sigma \approx 95.4\%$) were flagged as outliers. Given the relatively small dataset, these points were not discarded but prompted a re-examination of the experimental conditions for those specific runs.

B. CFD-modeling

1. Governing equations and numerical model

A complementary CFD study was conducted using ANSYS Fluent to supplement and extend the experimental data. The flow was modeled as adiabatic and incompressible. Since the flow remained stable even at the highest velocities, a steady-state formulation was employed, omitting the transient term from the Navier–Stokes equations

$$\nabla \cdot (\rho \mathbf{u}) = 0, \tag{2}$$

$$(\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \nabla R_{ij}, \tag{3}$$

$$R_{ij} = \nu_t (\nabla^2 \mathbf{u}) - \frac{2}{3} k \delta_{ij}. \tag{4}$$

The Reynolds stress term ∇R_{ij} was modeled using the k–kl–ω turbulence model proposed by Walters and Cokljat,[20] selected based on its successful application in similar confined particulate flows.[5]

Simple boundary conditions were applied: a uniform velocity profile U_s at the inlet, a zero-pressure gradient at the outlet, and no-slip conditions at all fluid–solid interfaces (pipe wall and particle surfaces).

2. Computational domain and mesh

The computational domain was a cylinder matching the experimental pipe dimensions: diameter 50 mm and length 1000 mm. Particles were arranged in a linear array and incorporated into the domain via Boolean subtraction, as shown in Fig. 3. The geometry was parameterized in ANSYS DesignModeler using the number of particles N_p and the center-to-center spacing δ as key inputs, allowing automatic rebuilding for each design combination. Smooth particles were modeled as perfect spheres; dimpled particles were geometrically reconstructed based on a standard 1.68-in. golf ball profile.

A mesh dependency analysis was performed on a domain with smooth particles by systematically reducing the average cell size Δx from 3.1 to 0.5 mm (Fig. 3). Critical output parameters—pressure drop and extrema of pressure and velocity fields—exhibited asymptotic convergence as $\Delta x \to 0$, with convergence achieved at $\Delta x = 1.3$ mm. This cell size was adopted for all subsequent simulations, resulting in meshes of approximately 0.1×10^6 cells for smooth particles and 1.2×10^6 cells for dimpled particles.

The fluid–solid interfaces were resolved with up to five prismatic boundary layers with a growth rate of 1.5, ensuring a dimensionless wall distance $y^+ \approx 1$ for the first layer. Although prismatic layer

FIG. 2. Connected particles with ABS separators and suction cups.

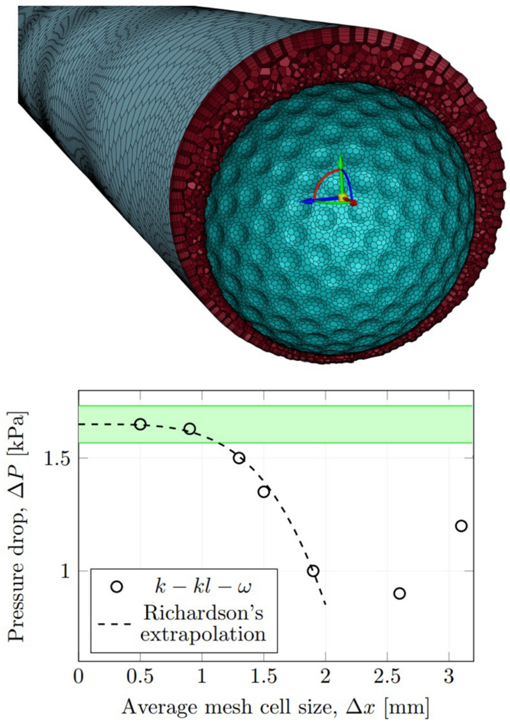

FIG. 3. Computational mesh (left) and mesh size dependency analysis (right) for $\delta_{\text{dimpled}} = 82$ mm.

stepping occurred in regions of extreme proximity (particle–particle and particle–wall), its impact was considered negligible due to the near-zero average velocities in those zones.

III. RESULTS AND DISCUSSION

The primary analysis relied on experimental data due to its superior reliability. First, smooth and dimpled particles were analyzed separately by examining the correlation between the normalized pressure drop and the superficial velocity. These two particle types were then compared within the same correlation framework. Two normalization methods for pressure drop were considered: per unit column length ($\Delta P/L$) and per particle ($\Delta P/N_p$). Finally, a correlation between the friction factor and the superficial velocity was established for both particle types.

CFD simulation results served a secondary role, primarily to validate the experimental findings. They also extended the available dataset by a factor of two, providing results over a wider range of superficial velocities. The resulting response surface from the CFD model integrated the key design parameters (inter-particle spacing δ and pressure drop ΔP), the primary operating parameter (superficial velocity U_s), and the model output—specifically, the ratio of pressure drops for dimpled to smooth particles.

A. Experimental data analysis

Figure 4 presents a comparative analysis of pressure drop per unit length ($\Delta P/L$) as a function of superficial velocity (U_s) for packed beds composed of smooth and dimpled spherical particles, across

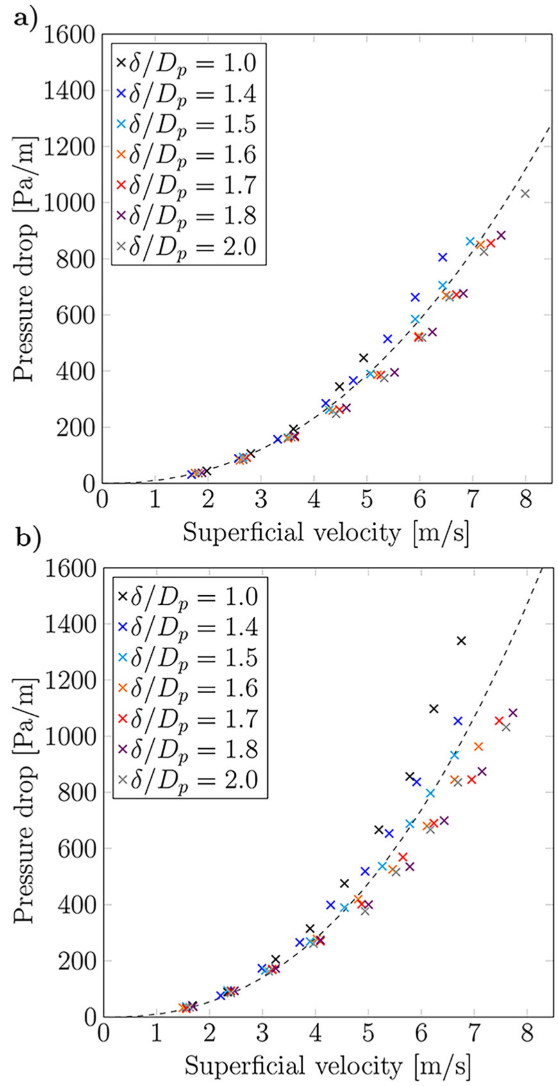

FIG. 4. Length-based normalization of the pressure drop for smooth (a) and dimpled (b) particles.

varying inter-particle spacing ratios (δ/D_p). The ratio δ/D_p represents the dimensionless center-to-center distance between neighboring spheres, where δ is the actual separation and D_p is the sphere diameter. These plots reveal distinct flow resistance behaviors influenced by both surface morphology and packing geometry. In this figure, experimental data points correspond to varying inter-particle spacing ratios (δ/D_p), where δ is the center-to-center distance between neighboring spheres and D_p is the sphere diameter. Dashed lines represent power-law approximations of the experimental data.

Figure 4 shows that for both particle types, a consistent and expected trend is observed: pressure drop increases with superficial velocity across all packing densities. This follows the fundamental Darcy–Forchheimer behavior, where pressure loss consists of viscous (linear at low *Re*) and inertial (quadratic at higher *Re*) contributions.

More significantly, for a given superficial velocity, the pressure drop decreases systematically as the inter-particle spacing ratio δ/D_p increases. This inverse relationship is physically intuitive, as a larger δ/D_p indicates a looser, more porous packing arrangement (lower solid fraction), resulting in reduced flow resistance and a lower pressure gradient for the same volumetric flow rate.

A direct comparison between the two plots reveals that dimpled particles consistently exhibit a higher pressure drop than their smooth counterparts at identical δ/D_p ratios and superficial velocities. For example, at a mid-range velocity of ~ 4 m/s and a packing ratio of $\delta/D_p = 1.6$, the dimpled particle bed shows a pressure drop of approximately 800 Pa/m, while the smooth particle bed under the same conditions shows only about 600 Pa/m. This represents an increase of roughly 33% for the whole dataset. At the same time, this increase varies across δ/D_p ratio.

This increased flow resistance for dimpled particles can be attributed to enhanced flow disturbance:

- Increased effective surface area: The surface dimples create a microscale texture that increases the total wetted surface area interacting with the air compared to a perfectly smooth sphere of the same nominal diameter.
- Promotion of flow separation and turbulence: The dimples act as surface imperfections that trip the boundary layer, promoting earlier flow separation and enhancing turbulent mixing in the inter-particle spaces. This increases viscous dissipation and form drag.

The sensitivity of pressure drop to changes in packing density (i.e., changes in δ/D_p) appears visually similar for both particle types. In both cases, the family of curves is well-separated, indicating that δ/D_p is a strong controlling parameter. However, a subtle difference may exist: the relative vertical spacing between curves for the dimpled particles appears slightly more compressed at higher velocities. This could suggest that for dimpled particles, the influence of bulk packing geometry (δ/D_p) becomes somewhat less dominant at high flow rates, where the surface-roughness-induced turbulence and drag dominate the overall pressure loss mechanism.

To quantify the effect of surface dimpling on flow resistance, the pressure drop per unit length for dimpled particles was normalized by that for smooth particles

$$\Psi = \frac{(\Delta P/L)_{\text{dimpled}}}{(\Delta P/L)_{\text{smooth}}}. \quad (5)$$

Figure 5 presents the ratio Ψ as a function of superficial velocity U_s for various inter-particle spacing ratios δ/D_p. Experimental data points are fitted with a smoothed dashed line to reveal trends across the tested velocity range. The results illustrate how both flow regime (U_s) and packing geometry (δ/D_p) influence the relative hydraulic performance of surface-textured particles in a confined monolayer arrangement.

Figure 5 shows that across all packing densities (δ/D_p), the pressure-drop ratio exhibits a consistent, non-monotonic relationship with superficial velocity. At low velocities (approximately 2–3 m/s), the ratio is relatively high, often exceeding 1.25. As velocity increases to a mid- and high-range (above 4–5 m/s), the ratio generally reaches a minimum, typically between 1.10 and 1.20. At higher velocities (above 6 m/s).

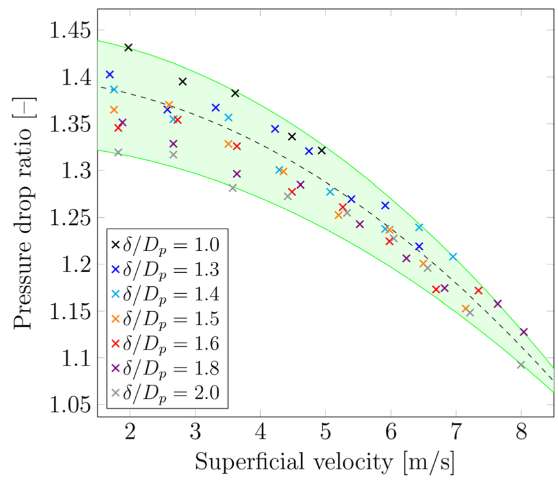

FIG. 5. Ratio of pressure drop per unit length for dimpled to smooth particles Ψ as a function of superficial velocity U_s for various inter-particle spacing ratios δ/D_p.

The observed decrease in the pressure drop ratio Ψ with increasing superficial velocity can be explained by considering the differential response of flow regimes and drag mechanisms between smooth and dimpled surfaces.

The key factor is the difference in the onset and progression of turbulence around each particle type. Smooth spheres exhibit a relatively delayed and sharper transition from laminar to turbulent flow in their boundary layer and wake. As velocity increases, smooth particles experience a relatively larger increase in pressure drop because: (i) delayed transition—the boundary layer on a smooth sphere remains laminar over a larger portion of the surface at moderate velocities. When transition eventually occurs (typically at a higher critical Reynolds number), it creates a sudden increase in skin friction and a broader, more energetic wake, leading to a sharp rise in form drag; (ii) turbulent wake growth—after transition, the turbulent wake behind a smooth sphere expands rapidly with increasing Reynolds number, creating greater base pressure deficit and higher overall drag.

In contrast, dimpled spheres exhibit fundamentally different flow characteristics: (i) early and controlled transition—the surface dimples act as distributed roughness elements that move the boundary layer, forcing transition to turbulence at a much lower Reynolds number. This results in a more gradual, less abrupt increase in drag with velocity; (ii) wake stabilization—the turbulence generated by dimples can lead to a more organized, re-attached flow pattern in the downstream separation region. This partially stabilizes the wake, reducing the rate at which the wake width and base pressure deficit grow with increasing velocity compared to a smooth sphere.

Figure 6 shows that for all velocities, Ψ decreases from approximately 1.41 to 1.08 as δ/D_p increases; at lower velocities (2–4 m/s), Ψ starts at higher values (1.3–1.41) and shows a steeper decline, while at higher velocities (6–8 m/s), Ψ starts at lower values (1.08–1.25) and decreases more gradually. These trends can be explained by the changing dominance of flow interaction mechanisms as particle separation increases.

At small δ/D_p (tight packing), the strong flow interference between neighboring particles reduces the effectiveness of dimples,

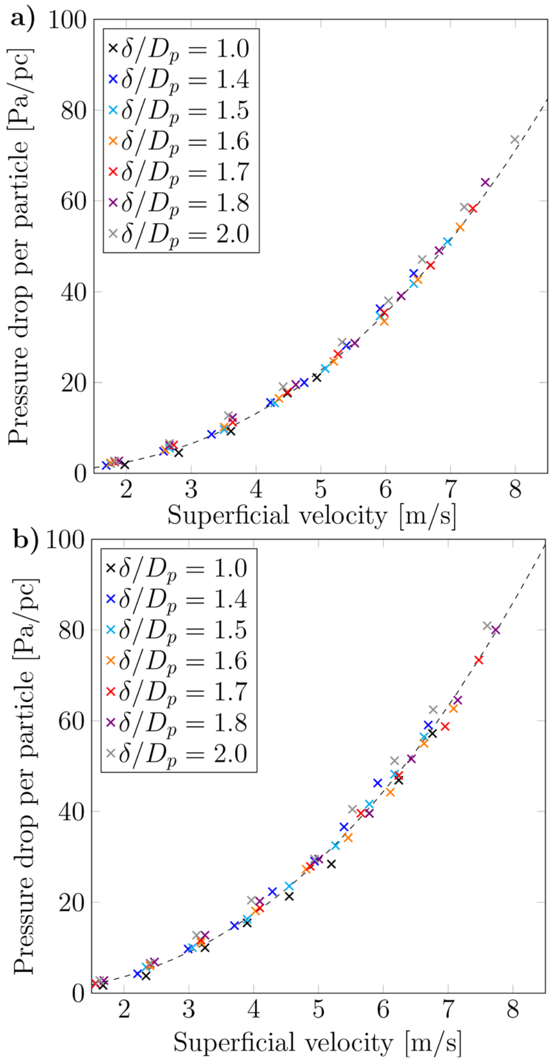

FIG. 6. Pressure drop per particle as a function of superficial velocity U_s for smooth (a) and dimpled (b) particles across various inter-particle spacing ratios δ/D_p.

since their primary aerodynamic benefit—early boundary layer transition and wake stabilization—is diminished when wakes from upstream particles directly impinge on downstream ones. The confined flow between particles limits the region where dimple-induced turbulence can fully develop.

As δ/D_p increases, particles become more isolated, allowing individual wakes to develop more completely. For smooth spheres, this results in larger, more energetic wakes that create significant form drag, while for dimpled spheres, the wake is already stabilized and narrower due to earlier transition. Consequently, the relative benefit of dimpling becomes more pronounced with greater separation.

Additionally, with larger spacing, each particle experiences a more developed incoming flow, which smooth spheres require for boundary layer transition—a process hindered in tight configurations. Dimpled spheres trigger transition earlier regardless of spacing, giving them a growing relative advantage as separation increases.

In tightly packed arrangements ($\delta/D_p \approx 1.0$), the high local velocities in narrow gaps affect both particle types similarly, masking specific dimple benefits. As spacing increases, flow becomes more uniform around each particle, allowing intrinsic aerodynamic differences to manifest more clearly.

During the experiment, the inter-particle separation distance δ determined the number of particles N_p within the fixed-length test section. Given the substantial contribution of each particle to the total pressure drop, an alternative normalization was examined: $\Delta P/N_p$. While both $\Delta P/L$ and $\Delta P/N_p$ follow power-law dependencies on superficial velocity U_s, the particle-based normalization shows markedly lower sensitivity to variations in inter-particle spacing δ/D_p, as demonstrated in Fig. 4. This collapse of the data indicates that pressure drop per particle remains approximately constant across different packing densities, simplifying the hydraulic characterization of such confined arrays.

Figure 6 presents the pressure drop per particle $\Delta P/N_p$ as a function of superficial velocity U_s for both smooth (a) and dimpled (b) particles across various inter-particle spacing ratios δ/D_p. A significant observation emerges from this normalization approach: when pressure drop is expressed on a per-particle basis, the experimental data for different δ/D_p ratios collapse remarkably well onto a single power-law trend for each particle type. This collapse indicates that the average pressure drop per particle is largely independent of spacing, though local variations may exist due to entrance and exit effects.

For smooth particles, all data points corresponding to δ/D_p values ranging from 1.0 to 2.0 align closely with the power-law approximation line, showing minimal scatter. The same behavior is observed for dimpled particles, where the experimental points for all spacing ratios follow a consistent trend with only minor deviations. This indicates that the pressure drop per particle $\Delta P/N_p$ is largely independent of inter-particle spacing δ/D_p and is primarily governed by the superficial velocity U_s and surface characteristics.

This collapse of data can be physically explained by considering that in a linear array configuration, each particle contributes approximately equally to the total pressure drop, regardless of its distance from neighbors. While the absolute pressure gradient $\Delta P/L$ decreases with increasing δ/D_p (as shown in Fig. 4), the number of particles N_p in a fixed-length test section simultaneously decreases. These two effects compensate each other, resulting in a nearly constant pressure drop per particle $\Delta P/N_p$ across different packing densities.

The power-law relationships can be expressed as follows:

$$\frac{\Delta P}{N_p} \sim U_s^b, \tag{6}$$

where the exponent b is approximately 2.34 for smooth particles and 2.31 for dimpled particles. The slightly lower exponent for dimpled particles suggests a less steep velocity dependence, consistent with their earlier boundary layer transition and more stable wake development across the velocity range.

This particle-based normalization reveals a fundamental scaling: in a confined linear array of spheres, the hydraulic resistance contributed by each individual particle remains remarkably consistent regardless of packing density. This finding simplifies the prediction of pressure drop in such systems, as the total pressure drop can be estimated by multiplying the single-particle contribution by the total number of particles, independent of their precise spacing arrangement.

It is well established that pressure drop in packed beds consists of two contributions: a linear (viscous) term and a quadratic (inertial) term. To quantify the influence of each term on the total pressure drop, a friction factor can be calculated. As shown in the seminal work by Nikuradse,[21] in the laminar regime, the friction factor decreases with increasing superficial velocity, in the transitional regime, it increases with increasing superficial velocity, and in the turbulent regime, it attains a nearly constant value as the superficial velocity rises.

The friction factor used here, defined as $f_s = \Delta P/(L \cdot U_s^2)$, quantifies the dimensionless flow resistance of the particle array. Figure 7 presents f_s as a function of superficial velocity U_s for both smooth and dimpled particles across a range of inter-particle spacing ratios δ/D_p.

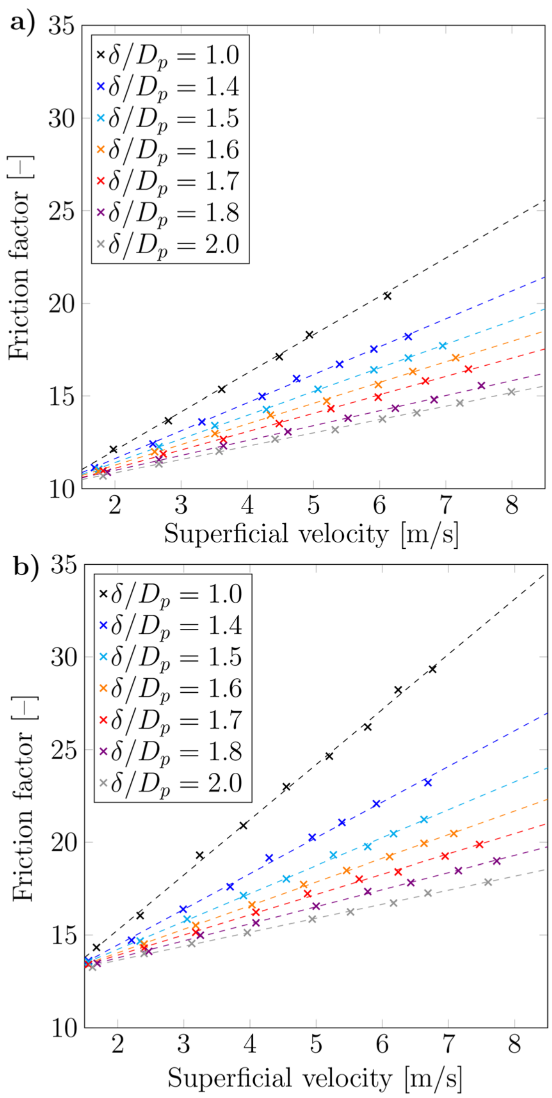

FIG. 7. Friction factor $f_s = \Delta P/(L \cdot U_s^2)$ as a function of superficial velocity U_s for (a) smooth and (b) dimpled particles across various inter-particle spacing ratios δ/D_p.

For both particle types, f_s decreases with increasing U_s, following a power-law trend, characteristic of porous media flows transitioning from a viscous-dominated regime at low velocities to an inertia-dominated regime at higher velocities. This behavior aligns with the Forchheimer extension of Darcy's law, where total pressure drop consists of both linear (viscous) and quadratic (inertial) contributions.

A clear separation between curves corresponding to different δ/D_p values is observed for both smooth and dimpled particles, indicating that packing density remains a significant parameter influencing dimensionless resistance. Smaller δ/D_p (tighter packing) yields higher friction factors due to increased flow constriction and more frequent flow redirection.

Comparing the two plots reveals that dimpled particles exhibit systematically higher friction factors than smooth particles at identical δ/D_p and U_s. This enhancement is consistent across all tested velocities and spacing ratios, confirming that surface dimpling increases effective flow resistance. The relative increase in f_s for dimpled particles is most pronounced at lower velocities and tighter packing, where viscous effects and surface-induced flow disturbances dominate.

The experimental setup shown in Fig. 1 is limited to an air flow velocity of up to 8 m/s through the monolayer. To extend the range of the experimental data, a series of numerical simulations was performed for higher velocities. The CFD model was validated against our experimental data for velocities below 8 m/s. For all datasets, excellent agreement was obtained between the numerical and experimental results, with a mean average error of less than 2%. The uniform inlet profile in simulations represents a simplification; however, the good agreement with experiments suggests the overall pressure drop is dominated by the strong flow constriction around particles rather than inlet details.

Contours of velocity and pressure for $\delta/D_p = 1.0$ and $\delta/D_p = 2.0$ are presented in Fig. 8.

Based on experimental data and CFD modeling results, the pressure drop ratio Ψ was determined for an extended range of superficial velocities. Figure 9 shows the pressure drop ratio as a function of both superficial velocity and inter-particle spacing ratio over this extended range. The figure clearly demonstrates that increasing the superficial velocity beyond 8 m/s leads to a further decrease in Ψ, reaching a

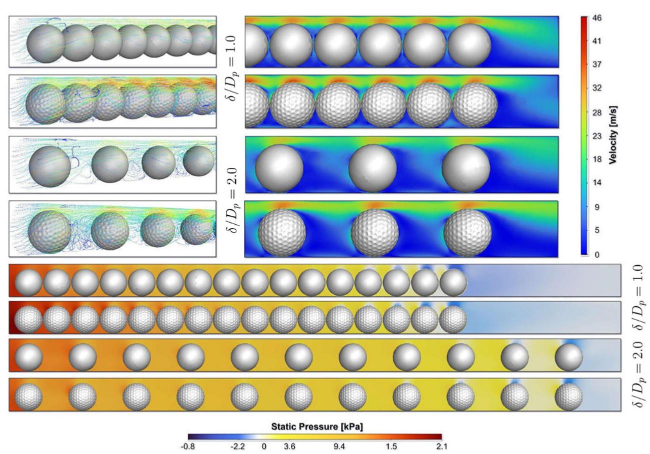

FIG. 8. CFD simulation results: flow visualizations (upper) and pressure distribution in lateral section (lower) for $U_s = 7$ m/s.

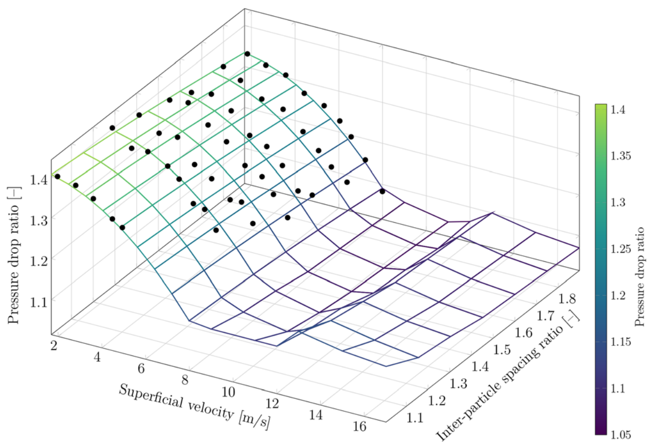

FIG. 9. Pressure drop ratio $\Psi = (\Delta P/L)_{\text{dimpled}}/(\Delta P/L)_{\text{smooth}}$ as a function of superficial velocity U_s and inter-particle spacing ratio δ/D_p for the extended velocity range (2–16 m/s) obtained through combined experimental measurements and CFD simulations.

minimum value of approximately 1.02 in the range of 9–10 m/s. A subsequent increase in superficial velocity above 10 m/s results in a rise of Ψ, achieving a local maximum near 14 m/s, beyond which Ψ gradually decreases again. Simultaneously, an increase in the inter-particle spacing ratio δ/D_p consistently leads to a slight reduction in the pressure drop ratio. This behavior can be explained as follows:

The observed trends can be interpreted through the behavior of the drag coefficient C_D as a function of Reynolds number (Re):[22,23]

- Smooth sphere: C_D experiences a sharp decline around $Re \approx 2 \times 10^5$ (the drag crisis) and remains relatively low at higher Re. In the transitional range relevant to these experiments ($Re \sim 10^4$–10^5), C_D for a smooth sphere remains relatively high and decreases gradually.
- Dimpled sphere: surface dimples eliminate the sharp drag crisis. C_D remains higher at low Re but decreases more gradually with increasing Re due to the early establishment of a turbulent boundary layer.

At a given velocity, the pressure drop ratio is approximately proportional to the ratio of the drag coefficients

$$\Psi = \frac{(\Delta P/L)_{\text{dimpled}}}{(\Delta P/L)_{\text{smooth}}} \propto \frac{(C_D)_{\text{dimpled}}}{(C_D)_{\text{smooth}}}. \tag{7}$$

As velocity increases and both particle types transition to higher Re, $(C_D)_{\text{smooth}}$ may decrease less rapidly than $(C_D)_{\text{dimpled}}$, causing their ratio—and hence Ψ—to decrease. This reflects the convergence of their drag characteristics as both flows become fully turbulent at high velocities.

The non-monotonic behavior of Ψ with U_s—specifically the local minimum near 9–10 m/s and a subsequent local maximum near 14 m/s—can be attributed to competing effects in the transitional flow regime. At velocities just beyond the experimental range (8–10 m/s), smooth particles may begin to approach their own transitional Reynolds number, experiencing a sharper drop in C_D relative to dimpled particles, thus reducing Ψ. As velocity increases further, wake interactions and compressibility effects (though slight) may temporarily amplify the relative drag of dimpled particles, leading to a local maximum in Ψ. Eventually, at even higher velocities, both flows are fully turbulent and Ψ stabilizes at a lower value. Meanwhile, the consistent decrease in Ψ with increasing δ/D_p indicates that the relative advantage of surface dimpling is diminished in more widely spaced configurations, where individual particle aerodynamics dominate over collective wake interactions. In addition, the computed flow shows strong acceleration (up to $5 \times U_s$) in the narrow gap above the sphere. The boundary layer remains thin and largely laminar over the accelerated portion but undergoes separation and turbulent mixing after the crest, especially for dimpled spheres. The friction Reynolds number is low [$Re_\tau \sim \mathcal{O}(10^2)$], indicating that the flow is not fully turbulent in the classical sense but experiences separation-induced mixing.

In summary, the complex dependence of the pressure drop ratio on both velocity and spacing stems from the distinct ways in which smooth and dimpled spheres transition to turbulence. Dimpled particles maintain a more gradual drag increase through controlled early turbulence, while smooth particles experience sharper transitions. This behavior is most pronounced in the transitional flow regime characteristic of packed-bed operations, where both viscous and inertial effects are significant.

IV. CONCLUSION

This work systematically examines the impact of surface dimpling on pressure drop in a confined monolayer of spheres, bridging the gap between classical aerodynamics of isolated dimpled spheres and practical engineering flows in tightly packed configurations. Through combined experiments and CFD simulations, it is shown that dimpled spheres increase flow resistance compared to smooth spheres across all tested inter-particle spacings and velocities. The pressure drop ratio Ψ decreases with increasing superficial velocity, exhibiting a minimum near 9–10 m/s, and is consistently lower for larger δ/D_p ratios. This behavior is explained by the distinct transition to turbulence: dimpled spheres trigger early, controlled boundary layer transition, leading to a more gradual drag increase, whereas smooth spheres experience sharper transitions and larger wake growth. Notably, pressure drop per particle ($\Delta P/N_p$) is found to be largely invariant with packing density, simplifying the hydraulic characterization of such arrays. The study highlights that in confined systems with low D/d ratios, the surface texture plays a decisive role in flow resistance, surpassing the influence of packing geometry at higher velocities. These results provide valuable guidance for the design of surfaces in applications involving narrow channels, packed beds, and confined particle flows, where wall effects and inter-particle interactions cannot be neglected.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Dmitry Pashchenko: Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Maxim Nikitin:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal);

Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

Abbreviations

CFD	Computational fluid dynamics
ID	Inner diameter
MAE	Mean average error

Greek symbols

Δx	Average cell size in mesh (mm)
δ	Center-to-center distances between particles (mm)
δ/D_p	Inter-particle spacing ratio
ν	Kinematic viscosity (m^2/s)
ρ	Density (kg/m^3)
Ψ	Pressure drop ratio for dimpled to smooth particles

Roman symbols

A	Fitting coefficient in pressure correlation ($Pa\,s^2/m^3$)
B	Fitting coefficient in pressure correlation ($Pa\,s/m^2$)
C_D	Drag coefficient
D_p	Particle diameter (mm)
D_{tube}	Tube inner diameter (mm)
f_s	Friction factor
k	Turbulent kinetic energy (m^2/s^2)
L	Length of test section (m)
N_p	Number of particles
R_{ij}	Reynolds stress tensor (m^2/s^2)
Re	Reynolds number
U_s	Superficial velocity (m/s)
$\mathbf{u}$	Velocity vector (m/s)
y^+	Dimensionless wall distance
ΔP	Pressure drop (Pa)
$\Delta P/L$	Pressure drop per unit length (Pa/m)
$\Delta P/N_p$	Pressure drop per particle (Pa)

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